Dynamic Lane Configuration for Improved Traffic Efficiency on Motorways

Majid Rostami-Shahrbabaki, Mehdi Keyvan-Ekbatani, *Member, IEEE*, Klaus Bogenberger, and Markos Papageorgiou, *Life Fellow, IEEE*

***Abstract*—The need for additional capacity in motorway networks during periods of high demand is unavoidable if congestion is to be prevented. Increasing capacity by building new roads is often infeasible, leaving operation-based traffic control measures as the primary approach to exploit the existing infrastructure. In this paper, the novel concept of dynamic lane configuration is introduced, which opens a new avenue in motorway traffic control that harnesses the infrastructure for traffic improvement. The lateral capacity of the existing motorway infrastructure is under-utilized due to lanes that are much wider than the vehicle's width. Dynamic lane configuration suggests that while current wide lanes ensure safety during high-speed driving, lower speed limits can be actively imposed during times of high traffic demand, allowing the lane width to be reduced, thanks to the reduced required lateral gap between vehicles at lower longitudinal speeds. By narrowing the lanes prior to congestion, it is possible to reclaim wasted space and add lanes to the road, leading to a dynamic capacity increase during the operation. This dynamic infrastructure layout with demand-responsive lane configuration during operation bridges the traffic management and infrastructure design. A model-based optimal control approach is developed to model the dynamic lane configuration and to define the times and locations of changing lane configuration. The proposed approach is tested in a simulation environment on two different motorway networks, each with a different configuration and demand profile. The promising results indicate the potential of the proposed approach in congestion mitigation and reducing travel time.**

***Index Terms*—Dynamic lane configuration, mixed traffic conditions, traffic control, motorway network.**

I. Introduction

MOTORWAYS serve as the economic backbone, connecting major cities within a country and across international borders, that facilitate the transportation of passengers and goods. However, road traffic congestion due to the limited capacity of motorway networks creates various problems, inducing travel delays. In literature and practice, there are two distinct facets of addressing capacity issues: transportation planning and operation. Transportation planning encompasses long-term impacts, with the primary goal being network capacity increase via infrastructure design and development. However, the construction of new infrastructure is costly and environmentally questionable [1]. In contrast, transportation operation focuses on the day-to-day management of existing transportation systems, with the immediate objective of exploiting the existing network infrastructure by ensuring smooth traffic flow, reducing congestion, and enhancing safety [2]. Transportation planning and traffic management, although both are aimed at improving traffic network supply, are currently run as separate and independent entities. In other words, transportation planning lacks real-time feedback and flexibility for adjustments based on current circumstances. Consequently, modern motorway networks' capacity is underutilized daily due to congestion, particularly during rush hours, i.e., when it is most urgently needed.

Nearly all current traffic control and management measures try to increase the efficiency and stability of traffic flow when the infrastructure and the traffic demand are fixed. Such measures can be categorized into road-based and vehicle-based measures [3]. The former is applied to all vehicles driving on the road and mainly includes approaches such as ramp metering [4], [5] and variable speed limit (VSL) systems [6], [7] or integration of both [8]. While these control measures offer significant benefits, they may not fully address heavily congested traffic conditions, demanding additional capacity. Vehicle-based measures that target individual or small groups of vehicles exploit Vehicle Automation and Communication Systems (VACS) for more innovative approaches [9]. In the last decade, the application of Connected and Automated Vehicles (CAVs) comprised the vast majority of cutting-edge research towards addressing traffic congestion. Various CAV functions can enhance traffic efficiency by platooning and increase the stability of the traffic flow by decreasing shock waves [10] through optimal lane flow distribution and controlling lane change behavior [11], [12]. However, the maximum effectiveness of such vehicle-based measures is typically achieved at relatively high CAV penetration rates [13]. Forecasts suggest that by 2030, less than 10% of vehicle fleets will feature higher levels of automation, with only 5% having no automation, while the majority will feature lower levels of automation with various advanced driver-assistance systems (ADAS), including cruise control and lane keeping assists [14]. Thus, traffic improvements relying solely on

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CAVs may not materialize in the immediate future; therefore, innovative approaches are needed to address traffic congestion in mixed traffic conditions.

To address the aforementioned challenges, the innovative concept of dynamic lane configuration (DLC) is introduced in this paper, which opens a new avenue in motorway traffic control, namely, an infrastructure-based control measure. DLC concept allows for flexible lane configuration, i.e., lane widths and numbers at each road section, that suggests the capacity of motorways to be dynamic and responsive to demand. In contrast to other traffic control measures that are applied to vehicles, DLC manipulates infrastructure to enhance capacity and improve traffic flow efficiency; therefore, it bridges traffic management and infrastructure design.

The concept of DLC as a novel infrastructure-based traffic control measure for mixed traffic conditions is utterly original, and to the best of our knowledge, no single work or piece of literature addresses this concept with the mentioned objectives. Nevertheless, it is worth pointing out some works that also utilize infrastructure manipulation during traffic operation and may be considered to some extent related to the proposed concept. While altering lane configurations specifically for traffic control has not been proposed, it is common practice to manage lane usage through strategies such as managed lanes, which aim to optimize traffic flow, regulate demand, and prioritize certain vehicle types or travel conditions [15]. One such approach is hard shoulder running (HSR), which adds one additional lane to the road and, thus, provides additional capacity during rush hours, has been investigated and is indeed in use in some countries (see [16] for a review). One study on German highways indicates capacity increases by 20% to 25%, with an annual reduction in congestion duration of up to 90%, demonstrating the effectiveness of HSR in improving traffic flow [17]. Similarly, a study in Italy found that HSR can increase the capacity of a two-lane highway by up to 35%, while it observed no significant changes in general safety conditions following HSR activation [18]. However, Karan and Lina showed that integrating HSR with the VSL strategy has led to a 29.73% reduction in collisions [19]. Despite the benefits of HSR, it defunctionalizes the role of the hard shoulders designed as a place for broken-down vehicles to move out of the way of traffic and to enable emergency vehicles to reach accident scenes efficiently. Another lane management approach is the use of reversible lanes, also known as tidal flow lane control, which aims to match the total road capacity with bidirectional traffic demand by leveraging unused minor-flow direction lanes to increase capacity in the major-flow direction [20]. The operation of reversible lanes can be determined using historical traffic data to match peak demand in each direction or can be adaptive based on prevailing traffic conditions [21]. Another study found that an adaptive reversible lane strategy can improve traffic flow by up to 40% compared to conventional time-based reversible lanes [22]. Macroscopic network flow analysis in [23] revealed that when origin-destination (OD) demand is highly asymmetric, travel delays can be reduced by up to 60%. This approach, however, is only practical with asymmetric demand and at locations where two opposite directions are not physically separated, limiting its applicability.

From a methodological perspective, it is essential to highlight that none of the mentioned approaches directly affect the driving behavior and traffic characteristics on the other lanes compared to DLC, as they basically focus on lane management without altering lane width. However, it is worth noting that DLC can potentially integrate elements of hard shoulder running and tidal flow lane control more efficiently. It is worth mentioning that, very recently, a flexible lane design approach to increase the capacity of isolated signalized intersections has been proposed, which simultaneously determines the optimal lane number, width, type, allocation, and signal settings [24].

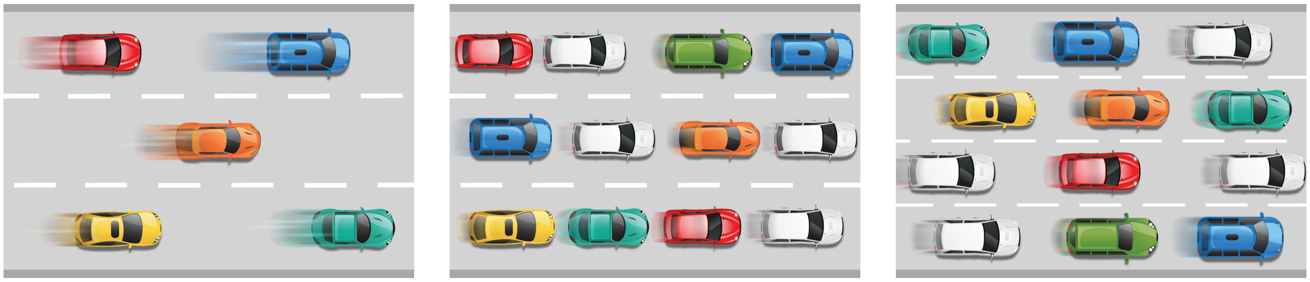
There are two other approaches to modifying the lanes in the era of CAVs: dedicated narrower lanes for CAVs [25] and lane-free traffic [26]. Indeed, the concept of lane-free traffic represents an advanced vision that increases road capacity [27], [28] but calls for a 100% penetration rate of CAVs, which may not be available in the coming decades, as noted previously. DLC, in contrast, proposes a pragmatic approach, specifically in countries where lane discipline is generally high, that retains traffic lanes while making them flexible and dynamic to enhance motorway capacity and tackle traffic congestion in the foreseeable future, where mixed vehicle fleets are the norm.

To evaluate the effect of this novel traffic control measure from the macroscopic perspective, an extended version of the Cell Transmission Model (CTM) [29] is developed to properly model traffic evolution in the presence of DLC. In addition, an optimal control approach is proposed for the optimal determination of the time and location of the motorway's segment where a new lane configuration should be applied. The developed approach is tested on two different motorway networks, each with a different configuration and demand profile. The promising simulation results indicate a significant improvement in terms of travel time and congestion mitigation.

The rest of the paper is organized as follows. Section II introduces the main concept of DLC in more detail. The developed methodology is elaborated in Section III. A proper simulation setup is developed in Section IV to test the proposed methodology, and the associated results are provided subsequently. Section V provides more insights into the challenges associated with DLC and our future works. The paper is finalized with the conclusion section.

II. Dynamic Lane Configuration Concept and Its Impact on the Motorway Traffic

Road design engineers employ wider lanes on motorways than in urban road networks to simultaneously consider traffic operation improvements and safety. The lane width depends on several factors, including the vehicle size, the required lateral clearance, the road type, and the operating speed [30]. The lane width on motorways usually ranges from 3.5 to 3.75 m, while a medium-sized vehicle has a width of about 1.9 m and a truck around 2.5 m [31]. Current fixed-width lanes lead to a huge loss of lateral capacity of the existing traffic infrastructure [26]. Thus, additional lateral occupancy on motorways could be realized if lane widths are reduced, assuming a proper traffic operation strategy is implemented. Therefore, we suggest that by actively imposing speed limits, the required lateral gap between vehicles can be reduced,



(a) Wide lanes for higher speed close to the desired speed

(b) Lateral capacity loss with wide lanes during congestion

(c) Narrow lanes allow additional lanes, providing extra capacity

Fig. 1. Dynamic lane configuration increases the road capacity to mitigate congestion.

allowing for narrower lanes and, thus, more lanes on the road. In other words, DLC suggests, for the first time, making the lateral gap between vehicles dynamic and speed-dependent, similar to the longitudinal gap. This action is intended for application in real-time, in response to the prevailing traffic conditions, leading to a dynamic and real-time adjustment of motorway capacity. This approach allows for capacity expansion during periods of high demand and a return to the original configuration during low-demand times, resulting in maximum throughput. This concept is illustrated in Figure 1.

While wide lanes accommodate significant lateral gaps between vehicles, as shown in Figure 1a, to enhance safety and comfort at high-speed driving, they become inefficient and lead to a loss of lateral capacity as traffic density increases and congestion ensues (Figure 1b). DLC suggests implementing a strategy that, prior to congestion, imposes a speed limit and then narrows down the lanes to a width that respects safety considerations and adds additional lane(s). The additional lane increases the capacity considerably as, in a similar approach, opening the hard shoulder for a two-lane road increases the capacity by up to 35% [18]. It is essential to emphasize that while implementing a speed limit and driving in narrower lanes indeed reduces capacity slightly, the capacity increment achieved by adding additional lane(s) is far more significant. Furthermore, supposing that this approach is not proactively implemented, congestion worsens as vehicles' speeds decrease naturally due to the increased demand, usually to a much lower value than the imposed speed limit. However, as will be shown in the following sections, the new lane configuration that requires a lower speed limit is usually downstream of the bottleneck areas. Therefore, by dynamically adding lanes, the throughput of the motorway increases substantially, and traffic congestion is mitigated or even avoided. This concept is not without precedent and is indeed already practiced in work zones where part of the motorway width is blocked due to construction, albeit not dynamically in real-time, but rather semi-permanently for the duration of the road works. For example, in Germany, regulations define a lane width of 2.6 meters at a speed of 80 km/h in work zones compared to the regular lane width of 3.75 meters [32].

This innovative infrastructure-based control approach that considers both the operation and design aspects of a traffic network proposes the dynamic capacity adjustment of motorways by dynamically changing the lane configuration spatiotemporally. In general, it proposes that the motorway stretch can be divided into several segments where the number of lanes changes dynamically at each segment, depending on the prevailing traffic conditions. Proposing multiple lane configurations along the motorway requires careful attention to the transitions between sections, particularly when the downstream section has fewer lanes. These practical considerations, along with lane-level vehicle control strategies, are further discussed in Section V.

This approach ensures that road infrastructure remains efficient and responsive to evolving transportation needs, featuring several benefits to traffic conditions and sustainable mobility as follows:

- **Efficient traffic and green transportation** Primarily, the dynamic capacity of motorways achieved through DLC approach significantly mitigates congestion, reducing associated adverse impacts such as travel time loss, fuel consumption, and emissions. DLC effectively addresses recurrent congestion, which occurs when demand exceeds nominal capacity, by providing additional capacity and non-recurrent congestion caused by random events like incidents or road works through dynamic lane reconfiguration. In addition, adopting multiple lane configurations can help distribute the wheel load more evenly across the pavement, significantly reducing the need for maintenance [33].
- **Traffic flow optimization** The golden rule of traffic flow optimization is trying to homogenize traffic flow with respect to time, location, and local speed differences [3]. DLC, by spatiotemporally changing the lane numbers equivalent to local capacity, provides smooth and homogenized traffic flow along the motorway. In addition, as the direct effect of the speed limit, the traffic flow will be homogenized with respect to the speed distribution, and the number of lane changes, a source of traffic flow interruption, will be reduced as an indirect effect. Empirical studies have confirmed that even a very low speed limit of 40 km/h provides a relatively high and stable flow of 1942 $veh/h/lane$ [34].
- **Improving safety** While traditional traffic engineers argue that wider lanes are safer, recent findings acknowledge that the road environment impacts human behavior. Narrower lanes result in less aggressive driving, less risky maneuvers, and an improved ability to slow down or stop a vehicle over a short distance to avoid collisions [35] and lead to fewer traffic injuries and fatalities [36]. In fact, very wide lanes may not necessarily enhance safety,

as the number of crashes on 2.8-meter lanes is similar to those on 3.4-meter lanes, indicating comparable safety outcomes [37]. However, some conflicting studies do suggest that narrower lanes may result in a 10% increase in crashes, or that adding extra lanes can reduce the number of crashes [38], [39]. However, these studies do not consider the impact of actively imposing speed limits, which are expected to increase safety [40]. In addition, the imposed speed limit included in the DLC concept is expected to decrease the number of lane changes, a risky maneuver known to be responsible for 10% of all accidents [41]. Furthermore, in addition to the contribution of lane width and speed limits, increasing capacity also plays an important role in ensuring safety. The dynamic capacity provided by DLC has the potential to prevent congestion; as a result, hazardous situations such as congestion shock waves and moving-jam stop-and-go waves, which can incur safety risks, can be avoided [42]. In fact, some studies indicated that the number of crashes reduces as the density of vehicles decreases [43]. More detailed safety assessment of DLC will be considered in the planned microscopic implementation.

- **Land use improvement** Providing additional capacity during peak traffic demand is an innovative approach. It allows for the design of narrower roads with fewer lanes that dynamically expand when and where needed, reducing the space and investment required for road construction. This approach optimizes land use and indirectly addresses environmental concerns associated with road construction.

In the next section, a methodology based on CTM is developed to model macroscopic traffic flow evolution in the case of DLC to prove and showcase the benefit of the introduced DLC concept. In addition, an optimal control approach is proposed to determine the optimal time and location where the lane configuration should be modified. The methodology is followed by simulation results and corresponding discussion.

III. Methodology

Since the proposed approach is novel and the microscopic and macroscopic behavior of traffic flow at sections with reduced lane width still needs to be investigated based on empirical data, we propose the use of CTM [29] with the triangular fundamental diagram whose parameters have a physical interpretation and thus are better predictable. Note that CTM is widely used in similar applications [44], [45], [46]. Figure 2 illustrates the fundamental diagrams for a given road section with original and modified lane configurations with λ and λ' number of lanes, respectively, where $\lambda \leq \lambda'$.

The jam density increases linearly with the increment of lane number to $\rho'_{max} = \epsilon \cdot \rho_{max}$, where $\epsilon = \lambda'/\lambda$. The variable $\sigma = v'_f/v_f$ is defined, where v'_f is the speed limit applied to the sections with narrower lanes and v_f is the free-flow speed in the original lane configuration. Note that ϵ is the independent decision variable, whereas σ depends on the speed limit v'_f to be employed in narrow-lane sections.

We initially assume that the backward wave speed does not change with the reduced lane width ($w' = w$); however,

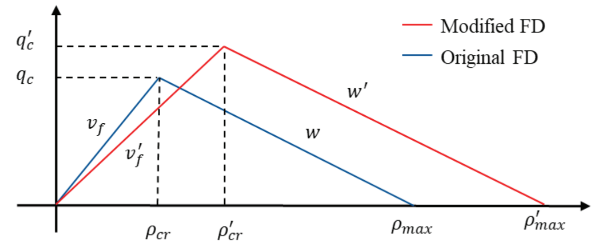


Fig. 2. Change in the fundamental diagram with dynamic lane configuration.

we are aware that it may change due to a different headway (possibly 10 $\sim$ 15% increase in headway [30]) and/or a slightly different reaction time of drivers at the reduced lane widths. This parameter can only be accurately determined based on empirical data. Therefore, we will select a range of wave speeds and conduct a sensitivity analysis to assess their effect on capacity improvement and the development of control decisions. Based on the fundamental diagram shown in Figure 2, the new capacity q'_c and the critical density ρ'_{cr} for the sections with modified lane configuration are calculated as:

$$q'_c = \frac{\epsilon w \sigma v_f}{\sigma v_f + w} \rho_{max} \tag{1a}$$

$$\rho'_{cr} = \frac{\epsilon w}{\sigma v_f + w} \rho_{max} \tag{1b}$$

Note that by replacing ϵ and σ with 1, we get the capacity q_c and critical density ρ_{cr} for the original lane configuration. In addition, Equations 1a and 1b show that the gained capacity and critical density are nonlinear functions of decision variables ϵ and σ.

A. CTM Modification With Capacity Drop Effect

Now, we write the CTM equations for each section i of the network, considering that $k = 0, 1, \ldots, K$ is the model time index. The conservation equation reads:

$$\rho_i(k+1) = \rho_i(k) + \frac{T}{L}\left[q_{i-1}(k) - q_i(k) + r_i(k) - \beta_i q_{i-1}(k)\right]$$
$$i = 1, \ldots, n \tag{2}$$

where ρ_i and q_i represent the density and outflow of section i, respectively, r_i is the on-ramp flow and β_i is the off-ramp exit rate. We assume that on- and off-ramps are located at the upstream end of the sections. In addition, n is the total number of sections, T is the model time step, and L is the section length. Without loss of generality, the network is assumed to be divided into homogeneous sections of the same length. The traffic flows are obtained as the minimum of the demand and supply functions, except for the last section, where we only consider the demand function. Different formulations are given in the literature for demand and supply function calculation. In this work, we use the extended version of CTM [46] that accounts for the capacity drop effect at on-ramps when merging congestion sets in. Furthermore, to reflect the DLC concept, the model must be adapted to account for changes in the number of lanes between consecutive sections. In fact, it is crucial to incorporate the capacity drop that may be caused by the reduction in the number of lanes to ensure that the model realistically captures the impacts on traffic flow and congestion that arise from such changes in lane configuration.

In the following equations, we assume a control time index $k_c = 0, 1, \ldots, K_c - 1$ associated with the control time step T_c. The model time step is typically set to 5-10 s for section lengths of 500 m, as it should comply with the so-called CFL condition. In contrast, the control time step should be higher than the model time step as enough time should be allowed for the vehicles to adjust their lateral location based on the new lane configuration before a new control signal arrives. Based on the above, CTM is extended as follows:

$$q_i(k) = min\left\{q_{D,i}(k,k_c), \alpha_i(k_c)\frac{q_{S,i+1}(k,k_c)}{1-\beta_{i+1}} - \theta_r r_{i+1}(k)\right\}, \quad i = 1,\ldots,n-1 \tag{3a}$$

$$q_n(k) = q_{D,n}(k,k_c). \tag{3b}$$

where the demand and supply functions are given by the following respective equations

$$q_{D,i}(k,k_c) = min\Big\{\sigma_i(k_c)v_f\rho_i(k), q'_c(k_c)\left(1+\theta_d\frac{\rho_i(k)-\rho'_{cr,i}(k_c)}{\rho'_{cr,i}(k_c)-\epsilon_i(k_c)\rho_{max}}\right)\Big\} \tag{4a}$$

$$q_{S,i}(k,k_c) = min\Big\{w\big(\epsilon_i(k_c)\rho_{max}-\rho_i(k)\big), q'_c(k_c)\Big\} \tag{4b}$$

Now, we elaborate on the additional terms in Equations (3) and (4) that account for the capacity drop effect. As explained in [46], in case of high on-ramp flow, $0 < \theta_r < 1$ imposes that the merge section $i+1$ can receive more flow than its nominal capacity. This allows for this section to become congested in case of sufficient demand at the upstream section i. Due to the fact that the density of the merge section can now be overcritical ($\rho_{i+1} > \rho_{cr,i+1}$), $0 < \theta_d$ leads to a decreasing demand function for this section, which imposes a smaller outflow, i.e., a capacity drop is reproduced. Note that in this work, we do not consider any ramp queuing and assume that the on-ramp flow is equal to its demand. Integration of ramp-metering with DLC will be considered in future works.

In a similar fashion, a new variable $\alpha_i(k_c)$ is introduced in this work to additionally reproduce the capacity drop due to lane reduction. In contrast to the two other constant parameters, α_i is a function of the control signal and, therefore, may take a different value for each section and control time-step.

$$\alpha_i(k_c) = max\left(1, 1+\theta_l\frac{\lambda_i(k_c)-\lambda_{i+1}(k_c)}{\lambda_i(k_c)}\right) \tag{5}$$

where θ_l is a non-negative tuning parameter smaller than 1.

This modification implies that if the number of lanes in the downstream section $i+1$ (λ_{i+1}) is less than the number of lanes in section i (λ_i), α_i becomes greater than 1, making a higher supply for section $i+1$. This may enable the flow of section i to be higher than its capacity, and thus, section $i+1$ may experience congestion. In such situations, given the role of θ_d in (4a), the flow of this section decreases, i.e. capacity drop materializes. In cases where the number of lanes does not change between the two consecutive sections or the downstream section has a higher number of lanes, α_i takes the value of 1, which does not change the supply function in (3a).

In summary, the combination of θ_r and θ_d and the combination of θ_l and θ_d are responsible for producing capacity drop due to the on-ramp and lane reduction, respectively. More specifically, if these parameters are set $\theta_r = 1$, $\theta_d = 0$ and $\theta_l = 0$, no capacity drop is reproduced, as typical for CTM; if these values are set between 0 and 1, the model produces a corresponding level of capacity drop.

B. Optimal Control Problem

The general macroscopic model explained in the previous section for motorway networks with DLC may be expressed in discrete-time state-space form as follows:

$$\mathbf{x}(k+1) = f\big[\mathbf{x}(k), \mathbf{u}(k_c), \mathbf{d}(k), \boldsymbol{\varphi}\big] \tag{6}$$

where $\mathbf{x}$ is the state vector comprising traffic densities at each section, $\mathbf{u}$ is the vector of control variables ϵ and σ for each section, $\mathbf{d}$ is the vector of external demand at the upstream-most section and the on-ramps, and $\boldsymbol{\varphi}$ is the vector of constant parameters, including fundamental diagram parameters v_f and w, and capacity-drop parameters θ_r, θ_d, and θ_l. While the state variables are real-valued, the control variables are integer by their nature. More specifically, we have the following constraints on the control variables:

$$\epsilon_i(k_c) \in \{1, \epsilon_{max}\} \tag{7a}$$

$$\sigma_i(k_c) \in \{\sigma_{min}, 1\} \tag{7b}$$

$$\big(\epsilon_i(k_c)-1\big)\big(\sigma_i(k_c)-\sigma_{min}\big) = 0 \tag{7c}$$

where ϵ_{max} is determined based on the maximum allowable number of lanes in the modified lane configuration, given the original lane configuration and the road width; σ_{min} depends on the imposed speed limit at the sections with the modified lane configuration. As mentioned earlier, the control variable σ depends on the lane configuration defined by ϵ. The additional hard constraint (7c) describes this dependency and states that if a modified lane configuration is implemented for a section, i.e., $\epsilon_i(k_c) > 1$, a corresponding speed limit should be applied, i.e., $\sigma_i(k_c) = \sigma_{min}$.

Based on the developed concept, the goal is to determine the time(s) and sections where the lane configuration should be changed. To this end, the objective function to be minimized is defined as follows:

$$J = T\sum_{k=1}^{K}\sum_{i=1}^{n}L_i\rho_i(k) + \phi_1\sum_{k_c=1}^{K_C-1}\sum_{i=1}^{n}\big(\epsilon_i(k_c)-\epsilon_i(k_c-1)\big)^2 + \phi_2\sum_{k_c=0}^{K_C-1}\sum_{i=2}^{n}\big(\epsilon_i(k_c)-\epsilon_{i-1}(k_c)\big)^2 \tag{8}$$

The first term in the objective function represents the TTS (total time spent). Minimization of TTS induces reducing the delay, thus increasing the traffic efficiency resulting from the actions of the proposed control measures. Since DLC is proposed for mixed traffic conditions, it is important for the practical implementation to minimize the number of lane configurations switching with respect to time and the neighboring sections. Therefore, the second and third quadratic terms are introduced in the objective function with the proper weighting factors to account for switching frequencies. Minimizing the first quadratic term penalizes the change in the lane configuration of each section in consecutive control time steps, while the second quadratic term penalizes the space variation of the lane configuration in two neighboring sections.

The optimal control problem (6)–(8) leads to a nonlinear and nonconvex mixed-integer programming that is hard to solve. Several investigations confirmed that conventional solvers are usually trapped in a local optimum in the vicinity of the initial solution. Therefore, a heuristic global optimization algorithm based on Particle Swarm Optimisation (PSO) is used to solve the optimal control problem. PSO is an optimization approach inspired by the movement of birds in search of food [47]. PSO conducts searches using a population of particles, each representing an individual. Each particle explores the search space for the optimal position (state) in a multidimensional continuous space. Each particle adjusts its position based on its speeds, which is influenced by its personal best position, which is the best position it has encountered so far, and the global best position, which is the best position found by any particle in the swarm. Since the decision variables in the developed model are integers, the binary version of PSO (BPSO) developed in [48] for improved convergence behavior is used.

To this end, a binary auxiliary decision variable $z_i(k_c) \in \{0, 1\}$ for each section is defined. $z_i(k_c) = 0$ represents the original lane configuration with $\epsilon, \sigma = 1$ while $z_i(k_c) = 1$ indicates modified lane configuration with $\epsilon = \lambda'/\lambda$ and $\sigma = v'_f/v_f$. BPSO is initialized with a population of m particles at random locations in the binary vector space $\mathbb{D} := \{0, 1\}^{n\times k_c}$. Thus, each particle $j = 1, \ldots, m$ is represented with a binary vector $z_m \in \mathbb{D}$. In addition, each particle has a speed vector $v_m \in \mathbb{R}^{n\times k_c}$ whose values for each dimension d at the iteration t are updated based on the following equation.

$$\begin{aligned} v_{m,d}(t+1) = c_0 v_{m,d}(t) + c_1 r_1 \big(p_{m,d} - z_{m,d}(t)\big) \\ + c_2 r_2 \big(g_d - z_{m,d}(t)\big) \end{aligned} \tag{9}$$

where c_0, c_1, and c_2 are weighting coefficients and r_1 and r_2 are random numbers between 0 and 1. p_m is referred to as the personal best, i.e., the location of the particle m with the best objective values in all iterations so far, and g is the general best, i.e., the best location among all particles so far. Note that personal and global best are found based on the objective function J. Once the particles' speeds are updated, the particles move to the new location in the solution space using the function $S(v_{m,d}) = |\tanh(v_{m.d})|$ as:

$$z_{m,d}(t+1) = \begin{cases} 1 - z_{m,d}(t), & r < S(v_{m,d}) \\ z_{m,d}(t), & r \geq S(v_{m,d}) \end{cases} \tag{10}$$

where r is a random number between 0 and 1.

The algorithm runs for a certain number of iterations until it converges to the optimal values of $\hat{\mathbf{z}} = \left[\hat{z}_i(k_c)\right]$ minimizing the objective function J, where the densities are obtained based on Equations (2)–(4). The convergence behavior of BPSO is demonstrated in the next section

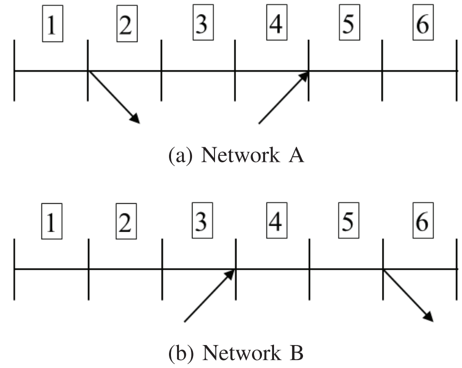


Fig. 3. Two networks considered for the evaluation of the DLC concept.

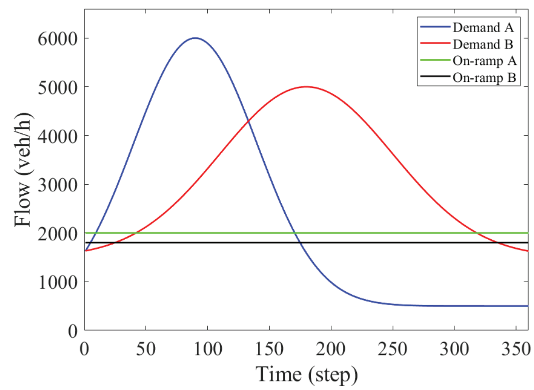


Fig. 4. Mainstream and on-ramp demand for the two networks.

IV. Simulation Results

A. Simulation Design

The DLC concept is implemented using the developed methodology and tested on two stretches of motorways with an initial configuration of 3 lanes. Each network is 3 km long and divided into 6 sections, but with a different configuration, as shown in Figure 3. Network A and its corresponding demand and simulation parameters are borrowed from [44]. The on-ramp in network B is placed before the off-ramp, creating a weaving section for a different demonstration of the effect of the DLC concept.

The 1-hour demand profile, i.e., mainstream and on-ramp demand, considered for both networks is shown in Figure 4. The exit rate of the off-ramps for the two networks is assumed to be 0.1. The modeling time step is 10 s, the control time step is 300 s, and capacity, free-flow speed, and backward wave speed for the CTM model are assumed to be 6000 veh/h, 100 km/h, and 12 km/h, respectively. In addition, the speed limit of 80 km/h is assumed for the sections with the modified lane configuration having 4 narrower lanes. The capacity drop parameters are set $\theta_r = 0.7$, $\theta_d = 0.4$ and $\theta_l = 0.4$, BPSO parameters are $c_0 = 1$, $c_1, c_2 = 2$, and weighting factors in the objective function are set to $10e{-3}$.

In the following, two cases are considered and simulated. In the first case, all lane configurations are fixed, whereas in the second case, the lane configurations are dynamic and change based on the defined optimal control problem.

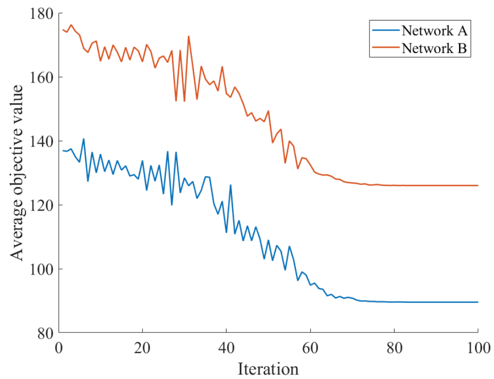


Fig. 5. Average objective value for each iteration of BPSO algorithm.

B. Analysis of DLC

We first illustrate the convergence of the BPSO algorithm. At each iteration, the objective value of each particle is stored. The average objective value across all particles at each iteration is calculated and presented in Figure 5 for both networks. Initially, the algorithm starts with a population of particles at random locations, analogous to random control signals, resulting in high objective values. As iterations progress, the algorithm explores better solutions and eventually converges to the optimal control signals with the minimum objective values. The figures demonstrate that after 80 iterations, the algorithm efficiently and swiftly converges to the best solution, highlighting the effectiveness and speed of the BPSO. The computational time for 100 iterations with 100 particles employed in MATLAB 2022b and executed on a personal computer with a processor Intel[1] Core[2] i7-8665U CPU@1.9 GHz and RAM of 32 GB was around 145 s.

Now, we demonstrate and compare the achieved results for network A. The relative densities are illustrated and compared for the two cases in Figure 6. Since each section with the dynamic lane configuration might have a different critical density, all densities are normalized with respect to the section's critical density, and thus, the relative densities are used for better comparisons. Consequently, the congestion is identified with the relative density being more than 1.

In the fixed lane configuration, as demand increases, congestion is triggered at the location of the on-ramp, which propagates upward. In addition, the merging cell 5 is also congested due to the capacity drop effect considered in the model. After 150 steps, when the mainstream demand reduces substantially, the congestion is mitigated and eventually dissolved, as demonstrated in Figure 6a. The relative densities for the dynamic lane configuration case are illustrated in Figure 6b. Comparing the two cases reveals the fact that the congestion is successfully suppressed by the implementation of DLC. To have a better understanding, the relative densities with the control signal and sections' outflows in two cases are shown and compared in Figure 7. It is important to note that the control signal is activated only once for sections V and VI. This means the new lane configuration is implemented only

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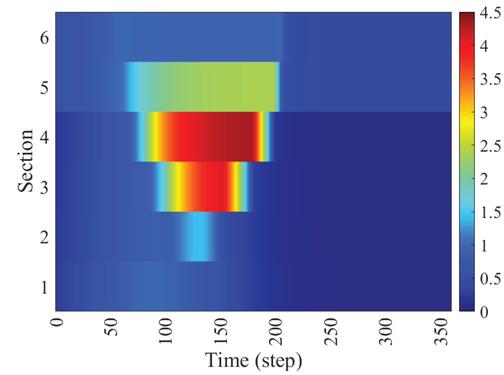
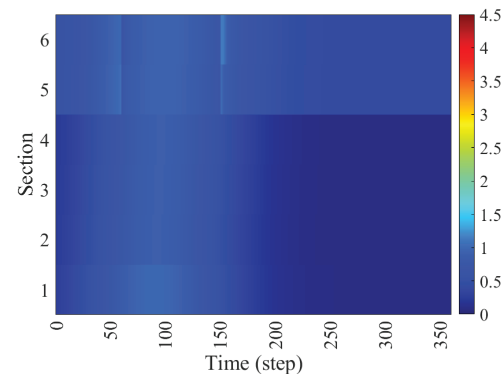


(a) Fixed lane configuration



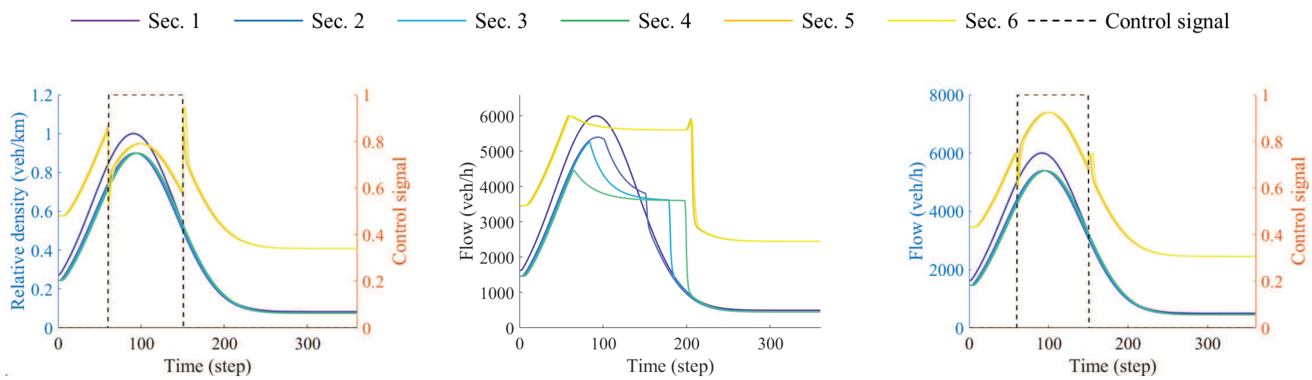
(b) Dynamic lane configuration

Fig. 6. Spatiotemporal relative density evolution in Network A in two cases.

for these two sections during high demand. This additional capacity provided downstream of the active bottleneck, i.e., downstream of the on-ramp location, keeps all the sections' relative densities below 1. This capacity increase is evident in the outflows as well. While the maximum capacity in case one, Figure 7b, is 6000 veh/h, the outflows of sections V and VI, Figure 7b, reach to 7391 veh/h highlighting a capacity increase of 23%.

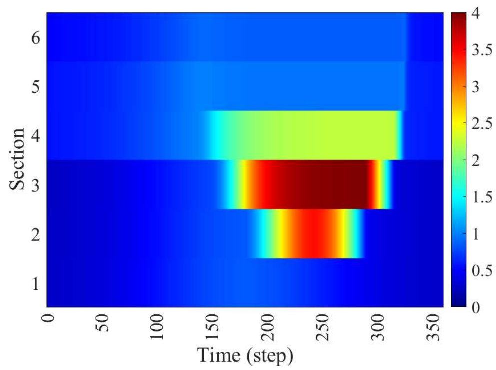
Spatiotemporal evolution of the relative densities for Network B are plotted in Figure 8. Similar to network A, the congestion starts at the on-ramp location, i.e., Section III, and continues as long as the demand at the mainstream is high. The merging section, i.e., Section IV, becomes congested due to the capacity drop effect in the model. Figure 8b indicates that with the DLC, the congestion is mitigated to a very high extent. The sections' relative densities are also shown in Figure 9. In contrast to Network A, the optimal controller generates two control signals for changing lane configurations in Network B. One is activated for 90 steps for sections IV and V, i.e., the weaving sections, and one for 30 steps for section VI. The former is needed to provide additional capacity for congestion mitigation, while the latter is generated to compensate for the capacity reduction induced by the reduction in lane numbers. We show in the next section that if the capacity drop term was not included in the model, the second control signal would not get activated.

In Network B, some sections still become congested. The investigation confirmed that an earlier activation and later deactivation of modified lane configuration for sections IV

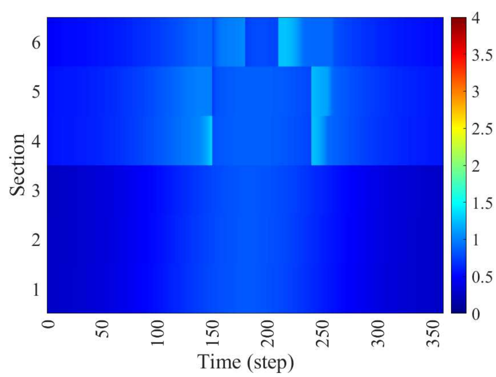


(a) Relative densities in dynamic lane configuration (b) Outflows in fixed lane configuration (c) Outflows in dynamic lane configuration

Fig. 7. Relative densities and outflows of Network A.



(a) Fixed lane configuration



(b) Dynamic lane configuration

Fig. 8. Spatiotemporal relative density evolution in Network B in two cases.

and V and a longer activation of the control signal for the last section would prevent such congestion, i.e., preventing the relative densities above 1 observed in Figure 9a. However, this led to higher TTS due to the longer period of the applied speed limit; therefore, based on the defined objective function, the obtained results are optimal. Comparing the outflows between the two cases also shows a capacity increase of about 13%. In addition, we carried out a further investigation by adding two more sections at the end of Network B and solved the optimal control problem again. The results indicated that the same control signal that was applied to section VI is now applied to sections 7 and 8 as well. Therefore, for the same time frame, the two new downstream sections also switch to more lanes.

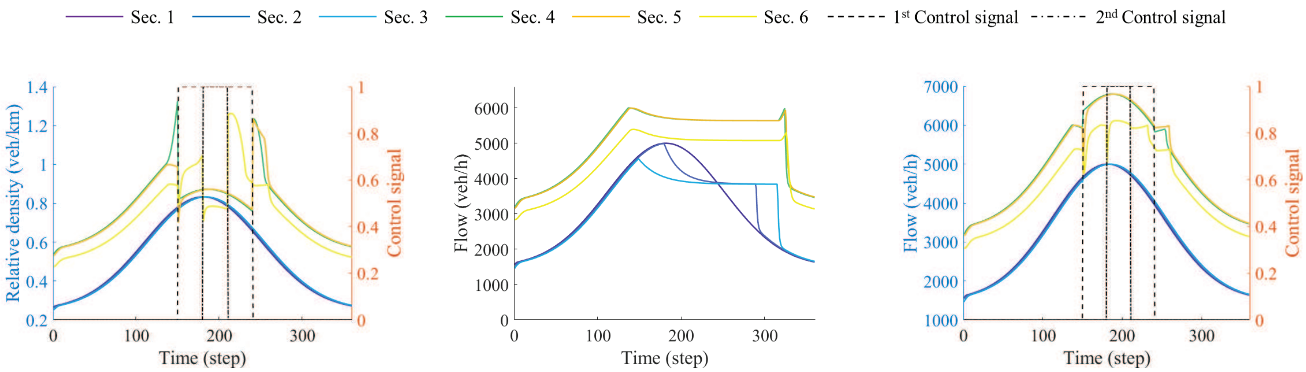
For Network A, the obtained TTS with a fixed lane configuration is 148 $veh \cdot h$, which is reduced to 89 $veh \cdot h$ with DLC, highlighting a 39% improvement. For Network B, DLC effectively leads to an improvement of 32% in TTS, reducing from 186 $veh \cdot h$ with fixed lane configuration to 126 $veh \cdot h$.

In summary, sections requiring a new lane configuration are identified by solving an optimal control problem. For both network topologies studied, the algorithm recommends modifying the lane configuration downstream of bottlenecks (triggered at on-ramp merge points in this study) to prevent congestion by ensuring sufficient downstream capacity for incoming demand. Thus, in Network A, sections V and VI switch to a modified configuration during high demand (see Figure 6b). Similarly, in Network B, sections IV and V shift to a four-lane configuration. However, unlike Network A, part of the demand in Network B exits via an off-ramp at the end of section V. As a result, section VI remains unchanged due to adequate capacity, unless the model accounts for capacity drops, in which case section VI also adopts the modified configuration; however, for a shorter time period (see Figures 8b and 10b. In future work, we will explore longer roadway networks and integrate additional strategies such as ramp metering; therefore, we expect a combination of different sections for applying modified lane configurations.

C. Analysis of Capacity Drop

As mentioned earlier, a modification of CTM is proposed to account for the capacity drop due to the number of lane reductions. In the results of the previous section, the capacity drop term is used in the model. In this section, we analyze its role in the model and its impact on the controller in detail. Since the controller in Network A suggests the change from three lanes to four lanes only for the last two sections, the lane reduction and, therefore, its impact on the capacity drop is not observed for this network. Therefore, we focus on Network B, where one lane drop is observed during the activation of the control signal, going from section V to section VI. To evaluate the effect of capacity drop introduced in Equations 3a and 5, three scenarios of dynamic lane configuration are considered for comparison:

- **Scenario 1** No capacity drop effect for the lane drop is considered. Dynamic lane configuration is applied based on the proposed optimal control approach.

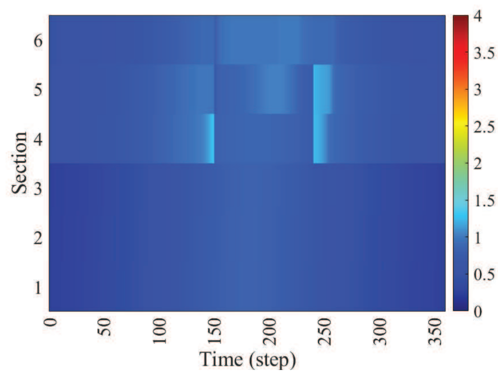


(a) Relative densities in dynamic lane configuration

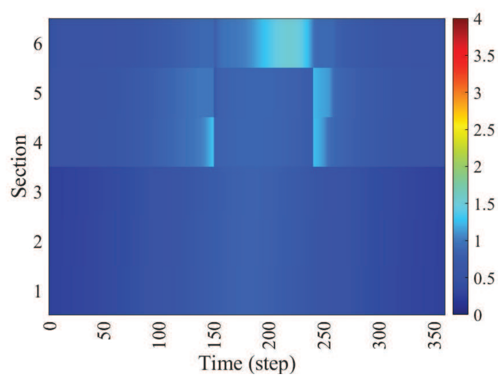
(b) Outflows in fixed lane configuration

(c) Outflows in dynamic lane configuration

Fig. 9. Relative densities and outflows of Network B.



(a) Scenario 1



(b) Scenario 2

Fig. 10. Comparing the effect of capacity drop on relative densities in Network B with DLC.

- **Scenario 2** The control signal in Scenario 1 is used, i.e., no capacity drop for controller design is considered. However, the capacity drop effect is considered in the model to evaluate the model under control.
- **Scenario 3** The model considers the capacity drop for a new controller design.

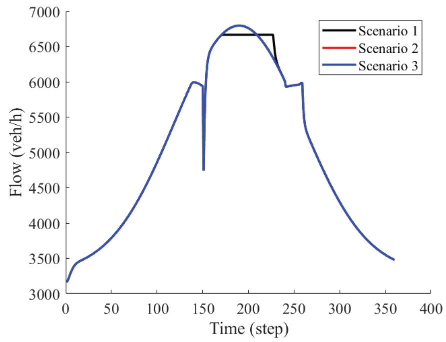
Note that Scenario 3 is the one that was already considered and discussed in the previous section. Therefore, the spatiotemporal relative density evolutions of the first two scenarios are shown and compared in Figure 10. Comparing figures 10a with 8a indicates that the DLC mitigates the congestion as expected, and only a little congestion is observed in section five during the activation of the control signal. However, the results may not be realistic since Section VI has one lane less than Section V during the activation of the control signal. Based on traffic flow theory, this phenomenon leads to a capacity reduction for this section. This fact materialized in Scenario 2, where Section VI becomes congested during the activation of the control signal in the upstream sections. In fact, once this capacity drop effect is considered in the controller, i.e., Scenario 3, the controller also suggests the lane modification for Section VI to compensate for this congestion, as shown in Figures 8b and 9a. An additional sensitivity analysis for the capacity drop parameter θ_l demonstrated that the results are insensitive to its value for this network topology and given demand pattern. For any value of θ_l greater than 0.1, we obtained the same control signal, with only minor variations in the flow values and corresponding TTS.

For a more detailed comparison, the flows and densities of sections V and VI in three scenarios are shown and compared in Figure 11. It is important to mention that the capacity drop introduced in the model affects the flows and densities of both sections, but the new control signal is only effective in the traffic condition of section VI. In Scenario 1, the density of section V increases as its outflow is limited by the capacity of the downstream section VI. With the capacity drop term being considered in Scenario 2, we allow the outflow of Section V to be higher than the nominal capacity of Section VI, which results in a lower density of Section V, shown in Figure 11a-11a. This additional flow to Section VI increases its density accordingly. With the new control signal for Section VI in Scenario 3, the additional lane allows for additional flow that partially compensates for the density increment in Scenario 2.

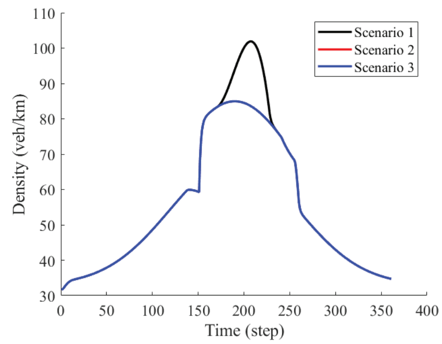
In addition, diverge locations at off-ramps can also be problematic and contribute to capacity reduction due to lane-changing behavior and the reduced speed of exiting vehicles upstream of the diverge. Analyzing this issue, along with other behavioral factors, and quantifying their consequences requires a microscopic implementation of the concept. This is one of the key paths of our ongoing research.

D. Sensitivity Analysis of the Model Parameters

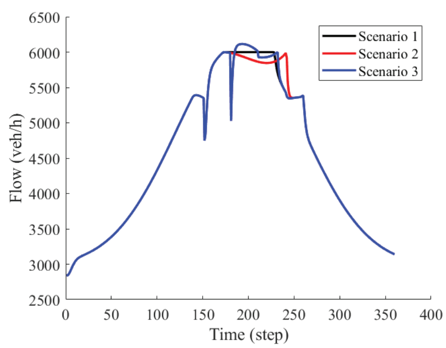
As mentioned earlier, microscopic driving behaviors, e.g., headway in the reduced lane width, and thus their macroscopic effects, such as backward wave speed, are yet unknown. In



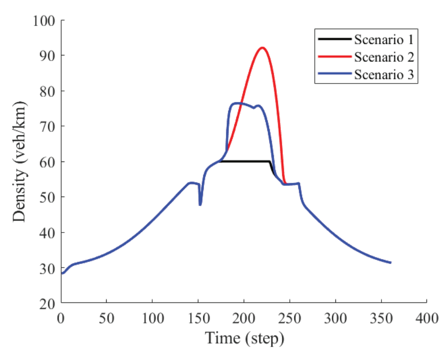
(a) Flow of Section 5



(b) Density of Section 5



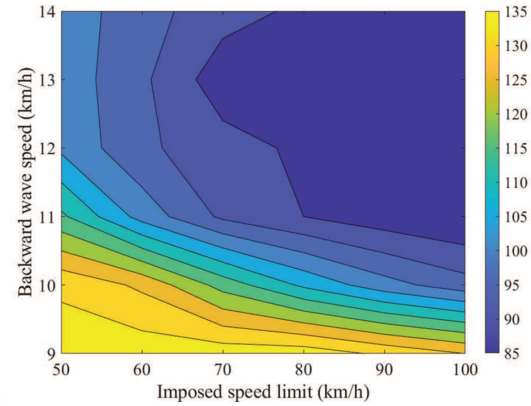
(c) Flow of Section 6



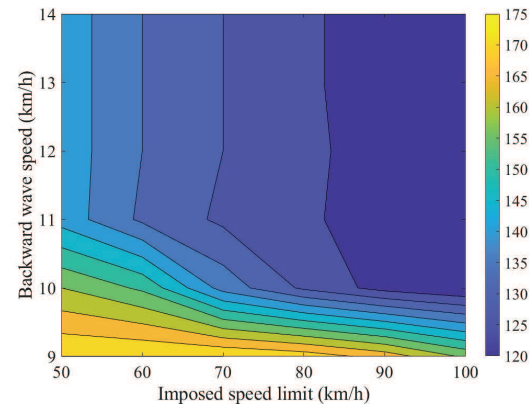
(d) Density of Section 6

Fig. 11. Comparing the effect of capacity drop on relative densities in Network B with DLC.

addition, a different regulation may apply to the speed limit for sections with narrower lanes. Therefore, we conducted a sensitivity analysis on the effect of different values of backward wave speed and speed limit on the achieved capacity



(a) Network A



(b) Network B

Fig. 12. Sensitivity of TTS with respect to the fundamental diagram parameters for both networks.

and traffic improvement. Figure 12 illustrates and compares the TTS for both networks based on different combinations of backward wave speeds and speed limits. Note that the capacity drop is assumed for all simulation runs, and the TTS for the fixed lane configuration assumed in the previous sections with $w = 12km/h$ and $v_f = 100km/h$ was 148 $veh \cdot h$ and 186 $veh \cdot h$ respectively. In addition, the basic settings used in the previous sections for DLC, i.e., $w' = 12\ km/h$ and $v'_f = 80km/h$, are also included.

The results indicate the nonlinearity of TTS with respect to both parameters, especially at higher values of w. A slower backward wave speed at sections with reduced lane width leads to lower capacity and, therefore, higher TTS. A 25% reduction in this speed leads to almost no improvement in the TTS. On the other hand, higher backward wave speed, associated with lower time gaps, results in higher capacity. This additional capacity exceeds the demands for the considered network and, therefore, does not impact the TTS considerably, at any imposed speed limit. This impact is more pronounced in Network B. Therefore, higher backward wave speed results in no change in control decision while lower ones, e.g., $w' = 10km/h$, lead to longer activation of control signals and/or applying modified lane configurations for more sections.

Regarding the speed limit, we may assume that different regulations may consider lower speed limits for the narrower lanes due to safety. However, we observe that even a very low speed limit of $50 km/h$ improves the TTS by 23.26%. Note that only lower speed limits did not change the control signal in both networks, as otherwise, it would have led to higher TTS values. A particularly interesting result emerged from the combination of low speed limits and low backward wave speeds. In this case, the control signal was implemented in the congested sections upstream of the bottleneck, unlike the typical scenario where downstream sections undergo a change in lane configuration. This behavior can be explained by the fact that, under such low values, the capacity of sections with modified lane configurations becomes lower than that of the original configuration (see Eq. 1a). As a result, limiting the capacity of upstream sections holds traffic further back, producing an effect similar to that of VSL. However, unlike VSL, the addition of an extra lane increases throughput, ultimately leading to a reduction in TTS. In addition, compared to the basic case, if the penetration rates of automated vehicles are high and thus the need for applying a speed limit is relaxed, even bigger improvement is achievable via more sections with modified lane configuration for longer times, which is comparable to the lane-free traffic with dynamic internal boundary [44].

V. Practical Considerations and Future Work

DLC, as an innovative traffic control measure, is introduced in this work. This concept requires careful consideration of various aspects, from modeling to control and implementation, which will shape our future research directions and are briefly discussed here.

One key area of focus is the evaluation of the impact of narrower lanes on driving behavior and their macroscopic consequence, which are yet largely unknown. To address this, traffic data acquisition is needed in sections with reduced lane width, such as work zones. Two potential data sources include drone-recorded videos and (extended) floating-car data (xFCD) of comparable work zones. Additionally, trajectories recorded from experiments conducted in driving simulators [49] can provide valuable insights into this problem. With such data, the microscopic behavior of drivers can be analyzed in more detail and be considered in new models. Current microscopic traffic models do not account for lane width, even though it may affect driving behavior. Therefore, we aim to develop and calibrate new microscopic models that incorporate lane width as a key factor.

Once the necessary traffic data for sections with narrower lanes is collected, macroscopic behavior, such as the fundamental diagram, can be more accurately estimated. This may lead to further development of macroscopic models or using second-order models such as METANET that model the speed dynamics, capable of more accurately reproducing the effect of speed limit on the traffic flow of the upstream section(s). Also, the optimal control problem developed in this work currently provides an open-loop solution. Future research will focus on creating a real-time feedback control system, potentially based on state feedback. Additionally, reducing the number of lanes upstream of a bottleneck can help to hold back traffic, achieving a similar effect to a VSL in mitigating congestion. Exploring the integration of this approach with VSL and ramp metering presents a promising avenue for future research.

Dynamic lane configuration changes on motorways also create new merge and diverge points, which must be carefully managed to avoid driving hassles during transition locations that can undermine the capacity improvements achieved by DLC. From a macroscopic perspective, if the traffic flow at the upstream and downstream sections of these points remains equal, shock waves and capacity drops can be avoided. The investigation presented in this paper, particularly concerning Network B, indicates that the controller maintains the control signal active for sections downstream of the bottleneck as long as additional capacity is required. In Network B, following the off-ramp where a portion of the demand is alleviated, the lane configuration reverts to its original state since the available capacity suffices to accommodate the demand. Otherwise, a modified lane configuration would have been proposed for section VI as well, similar to Network A. Consequently, transitions from a higher to a lower number of lanes occur only when the downstream capacity is adequate to handle the upstream demand, thereby rendering the merging situation manageable. In addition, mainstream flow control, often implemented through variable speed limits [34], is a key component of DLC. This could also be integrated into an optimal control problem to determine the ideal switching times to minimize the risk of shock waves. Nevertheless, we do not underestimate the challenges that may arise at the merging point, and thus, in the ongoing research by the authors, we are implementing DLC in a microscopic simulation for a more comprehensive analysis of such situations using lane-level control strategies, such as advisory systems for optimal lane flow distribution [50], [51]. Furthermore, CAVs' capabilities will be leveraged for managing the merging section, as several studies showed that even low penetration rates of CAVs can improve traffic conditions [52], [53].

As mentioned, DLC is designed to function effectively in mixed-traffic environments, accommodating both human-driven and (semi-)automated vehicles. Figure 13 illustrates a possible real-world implementation on a motorway stretch. For sections where increased capacity is needed, such as downstream of active bottlenecks, as demonstrated in this work, two-lane markings could be used. Information regarding new lane configurations and speed limits can be displayed via variable message signs or be communicated to vehicles from a roadside unit, either to be shown to the driver (via a head-up display) or be integrated into the vehicle's ADAS. A two-lane marking approach is currently being used in motorway work zones, where part of the road is blocked due to the construction. In such areas, vehicles are required to follow the new lane configuration, shown by yellow lane markings. Therefore, we suggest that two-color road markings could provide an immediate solution for the implementation of DLC as well; however, we acknowledge that this may be confusing for some drivers. Therefore, we envisage more advanced approaches, such as dynamic lane markings [54] or dynamic paint [55] using electrochromic paints [56], which may be

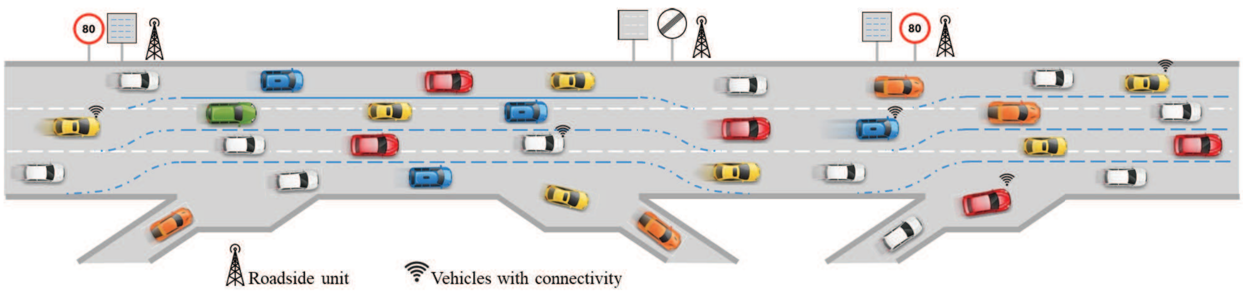


Fig. 13. DLC proposes multiple lane configurations in motorways in response to real-time demand.

considered in the future, allowing for a single lane marking to be present to prevent any confusion for vehicles. While lane changes may be needed to balance flows on different lanes and exploit new lateral space, it may be prohibited to ensure uninterrupted traffic flow for some sections, as shown in Figure 13 between lanes 3 and 4 in the weaving section. Note that in this research, without loss of generality, we assumed lane configuration from 3 to 4 lanes; however, the developed approach is generic and can be used for other lane configurations as well. In addition, lane configurations, particularly lane width adjustments, should be tailored to each country's specific regulations on motorway design and the composition of their vehicle fleet, especially considering differences in vehicle size and width. For example, a wider lane can be used on the right to be used by trucks or wider vehicles like Caravans. In addition to the points mentioned here regarding the feasibility of implementing DLC, we believe that since certain lane management approaches—such as hard shoulder running or tidal flow control—are already operational in some countries, the implementation of the DLC approach also appears feasible.

Evaluating innovative traffic control measures before real-world implementation requires the use of traffic simulation. Since DLC involves dynamically altering traffic networks in real-time, the development of a dedicated microscopic traffic simulator is necessary, as current simulators assume a fixed network topology. The simulator must be capable of handling real-time changes, such as adjusting lane numbers and widths. The microscopic implementation also allows for assessment of several lane-level features such as transition phase and safety metrics. We are in the course of developing a simulation framework, potentially based on the sub-lane feature of SUMO [57] or an extended version of it, which has been used for lane-free traffic [58].

In summary, while this concept is also applicable to manually driven vehicles, it offers greater benefits in mixed traffic conditions and can serve as a transitional phase toward a fully automated and connected environment, enabling more flexible approaches such as lane-free traffic. During this phase, the penetration rate of (semi-)automated vehicles is expected to increase over time, allowing the system to benefit from various levels of automation. Driver assistance systems, such as cruise control and lane-keeping, enable vehicles to operate effectively at reduced lane widths without compromising the capacity of road sections with modified lane configurations. As mentioned, even low penetration rates of (C)AVs can be helpful in managing the merge and diverge points introduced by DLC. A series of future studies is planned to explore various aspects of DLC, including modeling and control at both microscopic and macroscopic levels. We acknowledge that implementing DLC may require considerable time, as several aspects of this approach are still under investigation. However, such a gradual development process is typical for innovative traffic management strategies. Finally, instead of investing heavily in building new roads or expanding existing ones, we suggest prioritizing investment in upgrading current motorways to make them smarter and more adaptable. For instance, deploying adaptive lane marking technologies could accelerate the real-world implementation of DLC, increasing road capacity with flexibility in response to changing traffic conditions without the need for physical road extension. In future work, the integration and coordination of DLC with other control measures such as VSL and ramp metering for large-scale motorway networks will be explored.

VI. Conclusion

This paper introduced the novel concept of dynamic lane configuration (DLC), which proposes using lane adjustments as a traffic control measure on motorways. By dynamically changing the lane configuration, i.e., lane number and width of lanes at each motorway section, DLC enables adaptable capacity based on real-time traffic conditions. By manipulating the infrastructure during operation, DLC bridges transport planning and traffic management, offering additional capacity when needed and opening a new avenue for motorway traffic control. We employed the widely used CTM model with a triangular fundamental diagram to demonstrate the additional capacity gained via this novel approach. In addition, we modified the CTM further to account for the capacity drop due to the lane reduction, a situation that may occur with the implementation of DLC. Furthermore, an optimal control approach was proposed and solved with BPSO to determine the time and location where a new lane configuration should be applied. The proposed methodology was tested for two motorway networks, resulting in considerable congestion mitigation and reduced travel time. These two networks provide the necessary elements of a generic motorway, and consideration of a large-scale network is a part of our ongoing research. In addition, we performed a sensitivity analysis to evaluate the performance of the proposed algorithm with respect to changes in the fundamental diagram parameters. The outcomes indicated that a lower backward speed deteriorates the efficiency of the DLC approach while higher values provide additional capacity, allowing the network to accommodate higher demands. In addition, the total time spent is reduced almost linearly with the increase in the imposed speed limit. Even a very low speed limit improves the traffic condition considerably compared to the network with a fixed lane configuration. While the current work presents the first theoretical conceptualization of DLC, it serves as a foundational step toward a broader research agenda. In our planned series of future studies, outlined in detail in Section V, we aim to explore various dimensions of DLC. These efforts will contribute to the validation, refinement, and potential real-world implementation of the DLC approach in practical traffic environments.

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